COUNTABLE COLIMITS AS COKERNELS OF THE MAP 1 - F

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 10554-10566, following the statement at 10542-10552.

Editorial note. No omissions: both short exact sequences are displayed with their maps.

Allowed standard prerequisites (editorial).

- Exactness of N-indexed colimits. hypothesis of the lemma. Colim preserves the displayed short exact sequences.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Assume colimits over $\mathbf{N}$ exist and are exact. Then countable direct sums exists and are exact. Moreover, if (A_n, f_n) is a system over $\mathbf{N}$, then there is a short exact sequence*

$$0 \to \bigoplus A_n \to \bigoplus A_n \to \operatorname{colim} A_n \to 0$$

where the first map in degree n is given by $1 - f_n$.

Proof. The first statement follows from $\bigoplus A_n = \operatorname{colim}(A_1 \oplus \ldots \oplus A_n)$. For the second, note that for each n we have the short exact sequence

$$0 \to A_1 \oplus \ldots \oplus A_{n-1} \to A_1 \oplus \ldots \oplus A_n \to A_n \to 0$$

where the first map is given by the maps $1 - f_i$ and the second map is the sum of the transition maps. Take the colimit to get the sequence of the lemma. □

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